Taylor J Turner

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Robotics & Industrial Systems Software Engineer

Software and electrical controls engineer building APIs and middleware that connect robots, vision systems, and PLCs. Experience spanning logistics/warehouse automation, machine vision, and SCADA/MES—delivering production systems, CI/CD, and data collection with Python, C++, and .NET.

Skills

Languages	Python, C#, C++, TypeScript, T-SQL, Structured Text (IEC 61131-3)
Backend	FastAPI, ASP.NET Core, SQLAlchemy, Entity Framework, gRPC, REST
Frontend	Angular, React.js, HTML5, CSS3, (Tailwind CSS)
DevOps/Cloud	Azure, Docker, CI/CD (e.g., GitHub Actions/Azure DevOps), Containers, Blob Storage
Industrial/SCADA	PLCs (Studio 5000, TwinCAT, TIA Portal), Robots (FANUC), Vision (ifm), Ignition, InduSoft
Methods	OOP, Agile/Scrum, FMEA, Project & Product Testing, UL 508A

Work Experience

Bastian Solutions — Robotics *Software Engineer II (Apr 2025–Present); Software Engineer I (Sep 2022–Apr 2025)*

- Built software to coordinate robots, vision systems, and automation controllers for streamlined logistics operations (Python, C++, FastAPI, gRPC, Azure, Docker).
- Implemented cloud-backed storage and containerized deployments; streamlined CI/CD and reduced deployment time by **40%**.
- Productized R&D prototypes by adding testing frameworks and product analytics to support iterative improvements; stabilized product lines and enabled multiple contracted deployments.

Selected Projects

- **Pick and Drop** — Python app detecting pallet pocket locations; serialized motion data from ifm vision to a FANUC robot via socket comms.
- **Smart Pick** — Bin-picking pipeline integrating in-house vision and LLM guidance for motion; built annotation management with Azure Blob Storage and an Angular UI for filtering and AI dataset versioning.
- **Image Picker** — Screen controller syncing WMS picking orders with robots using OCR and HID mouse control to drive management software.
- **Stack Order** — Box-stacking pattern generator for palletization; migrated in-house tool from WinForms to .NET Core + React for multi-client web use.

Beckwood Press *Electrical Controls Engineer II (Oct 2021–Sep 2022); I (Oct 2016–Oct 2021)*

- Led electrical system design and PLC programming; designed UL 508A panels and drove successful customer acceptance testing.
- Built machine parameterization databases and standardized control programs to improve modularity and speed of deployment.
- Performed performance analyses on new product lines to inform enhancements and release readiness.

Selected Projects

- **Hydraulic Control** — Closed-loop position/force control with kinematic models; implemented feed-forward control, *Kalman* filtering, and target generation across multiple controller platforms.
- **Servo Press** — All-electric press using Yaskawa/Allen-Bradley servo controllers; encoder-based position and load-cell force feedback for highest-accuracy product line.
- **Auxiliary Systems** — Controls for custom infeed/outfeed, containment, and conveyance systems.

EPIC Inc. *Electrical Controls Technician (Dec 2014–Oct 2016)*

- Assembled electrical panels/machinery for pilot plants, vision systems, and custom industrial equipment; supported startup and factory acceptance testing to verify function and compliance (NEC, sanitary/wash-down, Class I Div I).

Education

Ranken Technical College — Associate Degree in Electrical Control Systems Dec 2014

Volunteer

Offensive/Defensive Line Football Coach, Titans of Troy, IL